HDOC - 65th day

TOWER OF HANOI USING RECURSION

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Problem Statement

Write a recursive C program to solve the Tower of Hanoi problem for n disks and display all moves.

The program must use recursion to move all disks from the source rod to the destination rod using an auxiliary rod.

The rules are: only one disk can be moved at a time, only the top disk of a rod can be moved, and a larger disk cannot be placed on a smaller disk.

Algorithm

1. Start.

2. Read the number of disks n.

3. Call towerOfHanoi(n, A, C, B), where A is source, B is auxiliary and C is destination.

4. If n = 1, move disk 1 directly from the source rod to the destination rod.

5. Otherwise, recursively move n - 1 disks from the source rod to the auxiliary rod.

6. Move disk n from the source rod to the destination rod.7. Recursively move n - 1 disks from the auxiliary rod to the destination rod.

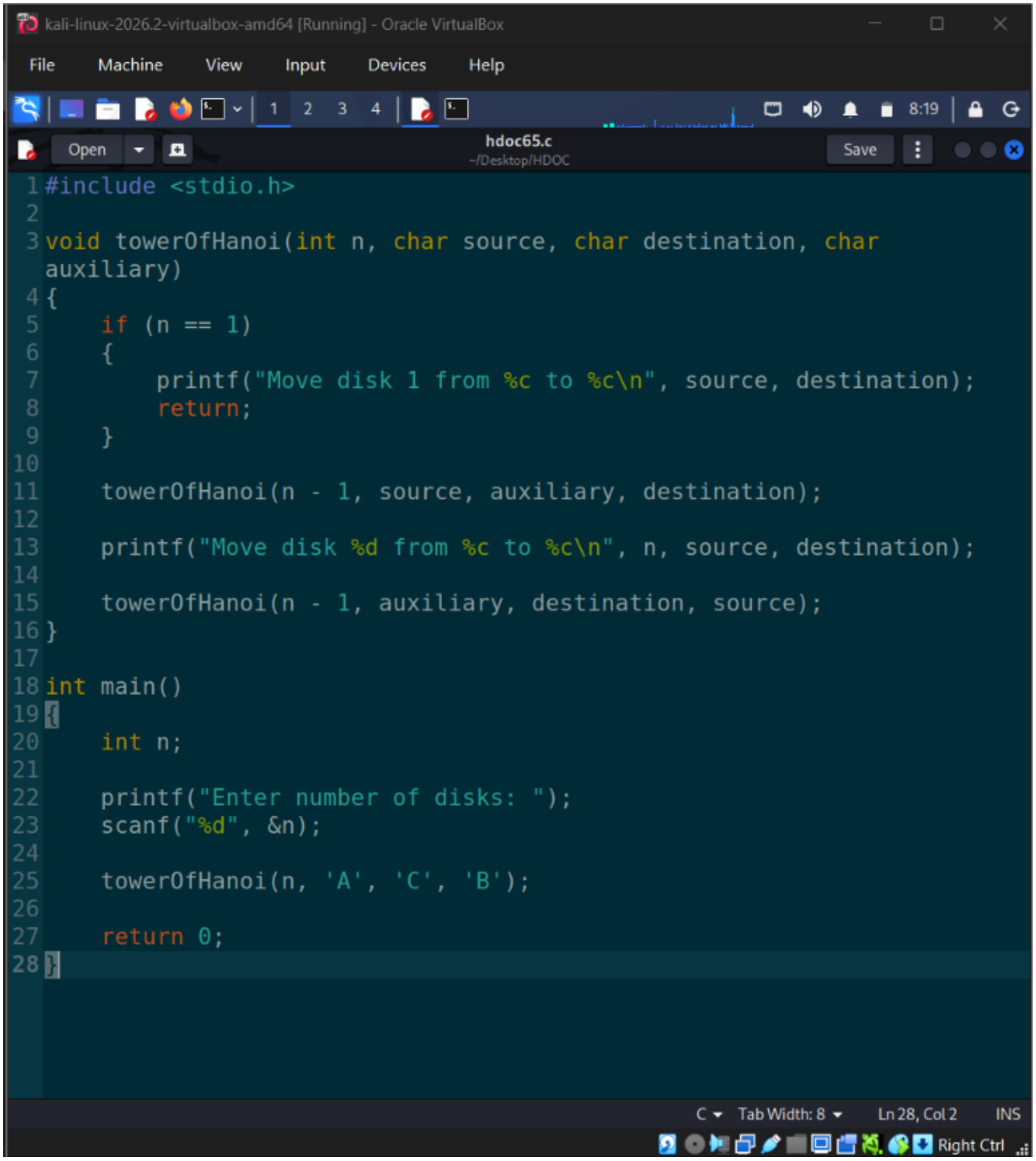
8. Display each move during the recursive process.

9. Stop.

Test Case Table

S. No.	Number of Disks (n)	Expected Output	Actual Output	Result
1	1	Move disk 1 from A to C	Move disk 1 from A to C	Pass
2	2	Move disk 1 from A to B Move disk 2 from A to C Move disk 1 from B to C	Move disk 1 from A to B Move disk 2 from A to C Move disk 1 from B to C	Pass
3	3	Move disk 1 from A to C Move disk 2 from A to B Move disk 1 from C to B Move disk 3 from A to C Move disk 1 from B to A Move disk 2 from B to C Move disk 1 from A to C	Move disk 1 from A to C Move disk 2 from A to B Move disk 1 from C to B Move disk 3 from A to C Move disk 1 from B to A Move disk 2 from B to C Move disk 1 from A to C	Pass

C Program


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
8:19
Open
hdoc65.c
~/Desktop/HDOC
Save
#include <stdio.h>
void towerOfHanoi(int n, char source, char destination, char auxiliary)
{
if (n == 1)
{
printf("Move disk 1 from %c to %c\n", source, destination);
return;
}
towerOfHanoi(n - 1, source, auxiliary, destination);
printf("Move disk %d from %c to %c\n", n, source, destination);
towerOfHanoi(n - 1, auxiliary, destination, source);
}
int main()
{
int n;
printf("Enter number of disks: ");
scanf("%d", &n);
towerOfHanoi(n, 'A', 'C', 'B');
return 0;
}
C Tab Width: 8 Ln 28, Col 2 INS
Right Ctrl

Compilation and Execution

Compilation command: gcc hdoc65.c -o hdoc65

```
┌──(kali㉿kali)-[~/Desktop/HDOC]
└─$ gcc hdoc65.c -o hdoc65
```

Execution command: ./hdoc65

Execution Screenshot:

